

FULL LIST OF PUBLICATIONS

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Journal Papers: SCI-Indexed

1. L. Geng, J. Shen*, **F. Zhang**. "Dynamic current logic based flip-flop design for robust and low power security ICs," *Electronics Letters*, 2017 [accepted](#)
2. X. Zhao, **F. Zhang***, S. Guo, Z. Gong. "Optimal Model Search for Hardware-Trojan-based Bit-level Fault Attacks on Block Ciphers," *SCIENCE CHINA Information Sciences*, 2017 [accepted](#)
3. P. Zhou, T. Wang, X. Lou, X. Zhao, **F. Zhang***, S. Guo. "Efficient flush-reload cache attack on scalar multiplication based signature algorithm," *SCIENCE CHINA Information Sciences*, 2017 [accepted](#)
4. **F. Zhang**, X. Zhao, W. He*, S. Bhasin, S. Guo. "Low-cost design of stealthy hardware trojan for bit-level fault attacks on block ciphers," *SCIENCE CHINA Information Sciences*, 2017 [accepted](#)
5. L. Geng, **F. Zhang***, J. Shen, X. Zhao, W. He, S. Bhasin, S. Guo. "Transistor level SCA-resistant scheme based on fluctuating power logic," *SCIENCE CHINA Information Sciences*, 2017 [accepted](#)
6. M. Khan, H. Zhang, **F. Zhang**, S. Shahzad, R. Ullah, S. Ali, Q. Arain, M. Ahmed "X-Band Power Amplifier for Next Generation Networks based on MESFET," *China Communications*, Vol.14, No.4, pp.11-19, 2017
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8. **F. Zhang***, S. Guo, X. Zhao, T. Wang, J. Yang, F. -X. Standaert, and D. Gu. "A Framework for the Analysis and Evaluation of Algebraic Fault Attacks on Lightweight Block Ciphers," *IEEE Transactions on Information Forensics and Security*, Vol.11, No.5, pp.1039-1054, 2016
9. **F. Zhang**, X. Zhao, S. Guo, J. Shen*, J. Huang, and Z. Hu. "A Comprehensive Study of Algebraic Fault Analysis on PRINCE," *China Communications*, Vol.12, No.7, pp.127-141, 2015

10. P. Zhou, T. Wang, G. Li, **F. Zhang***, and X. Zhao. "Analysis on the Parameter Selection Method for FLUSH+RELOAD based Cache Timing Attack on RSA," *China Communications*, Vol.12, No.6, pp.33-45, 2015
11. S. Guo, X. Zhao, **F. Zhang***, T. Wang, Z. Shi, F. -X. Standaert, and C. Ma. "Exploiting the Incomplete Diffusion Feature: A Specialized Analytical Side-Channel Attack Against the AES and Its Application to Microcontroller Implementations," *IEEE Transactions on Information Forensics and Security*, Vol.9, No.6, pp.999-1014, 2014
12. X. Zhao, S. Guo, **F. Zhang***, T. Wang, Z. Shi, Z. Liu, and J. Gallais. "A comprehensive study of multiple deductions-based algebraic trace driven cache attacks on AES," *Elsevier Computers & Security*, Vol.39, Part.B, pp.173-189, 2013
13. X. Zhao, S. Guo, **F. Zhang***, T. Wang, Z. Shi, H. Liu, K. Ji, and J. Huang. "Efficient Hamming Weight based Side-Channel Cube Attacks on PRESENT," *Journal of Systems and Software*, Vol.86, No.3, pp.728-743, 2013
14. X. Zhao, S. Guo, **F. Zhang***, T. Wang, Z. Shi, and H. Luo, "Enhanced Side-Channel Cube Attacks on PRESENT," *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences*. Vol.E96-A, No.1, pp.332-339, 2013

Conference Papers

1. **F. Zhang**, L. Geng, J. Shen, S. Bhasin, X. Zhao, S. Guo. "Improved low-entropy masking scheme for LED with mitigation against correlation-enhanced collision attacks," In *AsianHOST 2017*, [accepted](#)
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3. **F. Zhang**, X. Zhao, S. Guo, T. Wang, Z. Shi. "Improved Algebraic Fault Analysis: A Case Study on Piccolo and Applications to Other Lightweight Block Ciphers," In *COSADE 2013*, pp.62-79, 2013
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5. X. Zhao, **F. Zhang**, T. Wang, S. Guo, Z. Shi, H. Liu, K. Ji. "MDASCA: An Enhanced Algebraic Side-Channel Attack for Error Tolerance and New Leakage Model Exploitation," In *COSADE 2012*, pp.231-248, 2012. (**Best Paper Award**)
6. **F. Zhang**, Z. Shi, "Differential and Correlation Power Analysis Attacks on HMAC-Whirlpool," In *ITNG 2011*, pp.359-365, 2011.
7. S. Wang, **F. Zhang**, J. Dai, Z. Shi, L. Wang, "Making Register File Resistant to Power Analysis Attacks," In *ICCD 2008*, pp.577-582, 2008.
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9. **F. Zhang**, Z. Shi, B. Wang, "Chord-based Key Establishment Schemes for Sensor Networks," In *ITNG 2008*, pp.731-737, 2008.
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2. H. Chen, T. Wang, **F. Zhang**^{*}, X. Zhao, "Fault-based guess-and-determine attack on SOSEMANUK," *Journal of Huazhong University of Science and Technology (Natural Science Edition)*, Vol.45, No.2, pp.72-77, 2017. In Chinese.
3. J. Huang, X. Zhao^{*}, **F. Zhang**, S. Guo, P. Zhou, J. Yang, "Improvement and Evaluation for Algebraic Fault Attacks on PRESENT," *Journal on Communications*, Vol.37, No.8, pp.144-156, 2016. In Chinese.
4. H. Gan^{*}, H. Zhang, J. Li, **F. Zhang**, X. Zhao, P. He, "Independent component analysis applied in electromagnetic attack," *Chinese Journal of Radio Science*, Vol.31, No.2, pp.401-405, 2016. In Chinese.
5. H. Gan, H. Zhang, **F. Zhang**^{*}, X. Zhao, P. He, "Electromagnetic Information Leakage Acquisition and Pretreatment," *Chinese Journal of Radio Science*, Vol.30, No.5, pp.1004-1008, 2015. In Chinese.
6. H. Zhang, J. Li, **F. Zhang**^{*}, H. Gan, P. He, "A Study on Template Attack of Chip Base on Side Channel Power Leakage," *Chinese Journal of Radio Science*, Vol.30, No.5, pp.987-992, 2015. In Chinese.
7. H. Chen, T. Wang, **F. Zhang**^{*}, X. Zhao, Y. Sun, "Research on Algebraic Fault Analysis on HIGHT," *Journal of Shanghai Jiaotong University*, Vol.49, No.12, pp.1817-1825, 2015. In Chinese.
8. X. Wang, S. Guo, X. Zhao, M. Song, **F. Zhang**, "Research of power preprocessing optimization-based template attack on LED," *Journal on Communications*, Vol.35, No.3, pp.157-167, 2014. In Chinese.
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13. **F. Zhang**, S. Zhang, Y. Tang, W. Dai, "Design on Smart card-based Secure Logon System," *Information Security and Communications Privacy*, 2004(3):38-41. In Chinese.

Books and Chapters

1. Z. Shi, B. Wang, **F. Zhang**, "CBKE: Chord-based Key Establishment Schemes for Wireless Sensor Networks," *Chapter in Handbook on Sensor Networks*, (Xiao, Chen and Li, eds.), pp. 399-418, World Scientific Publishing Company, August 2010

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1. **F. Zhang**, Ph.D Thesis, "Towards Comprehensive Countermeasures to Power Analysis Attacks," December 2011.
2. **F. Zhang**, Master Thesis, "Identity Authentication and Access Control Based On Smart Card," January 2004. In Chinese.